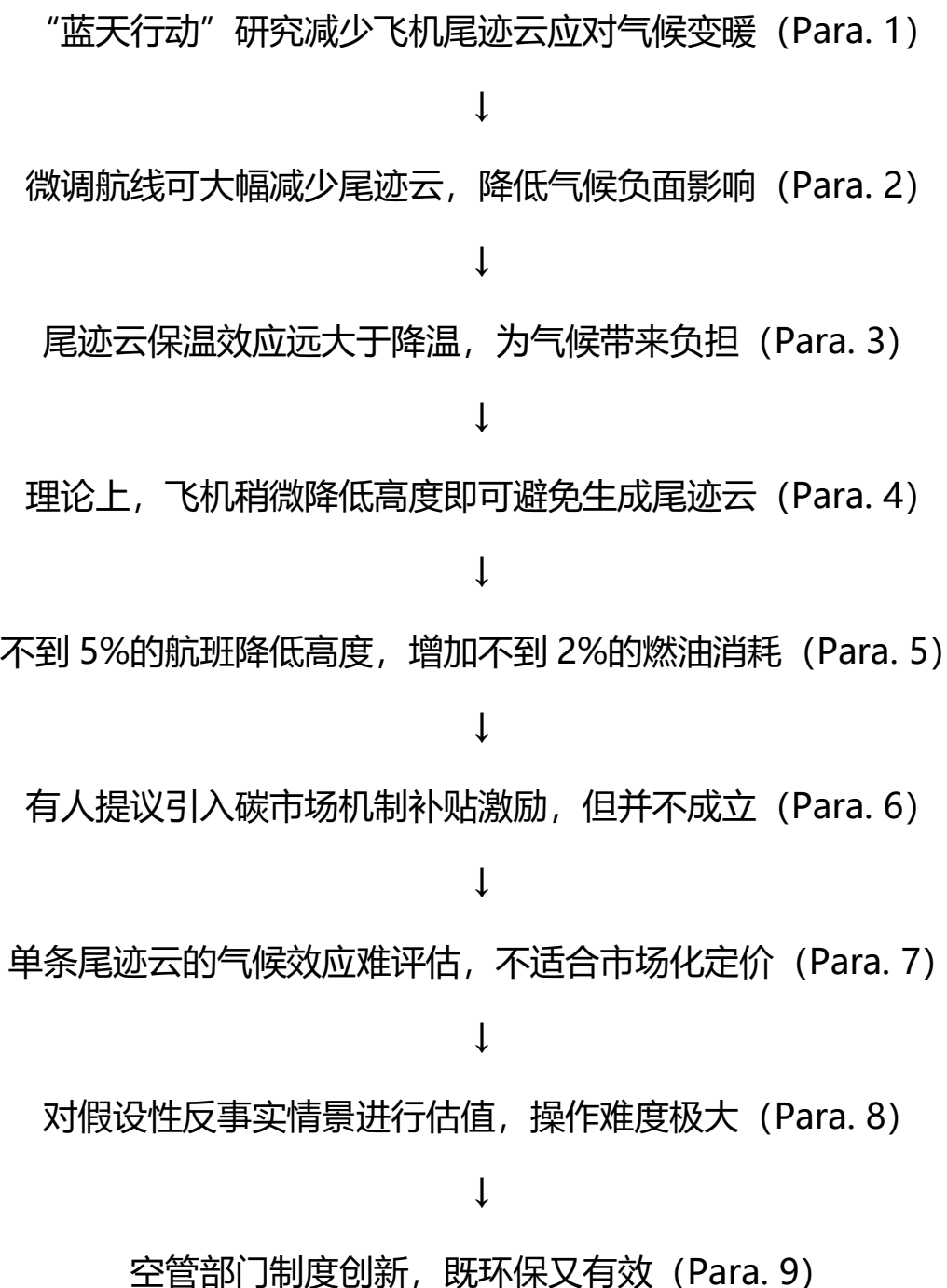


【8.30| Economist】 Without a trace 污影消痕

难度评级: ★★★☆☆

【考研那些事儿之科技系列】

今日文章脉络:



Fewer **contrails** would be good for the climate

飞机尾迹云越少，对气候越好

【学习流程：第一遍请先看原文自己精读，第二遍对照着本精读文档进行理解。文章标注的单词为重点单词，只提供相关注释，想要详细释义麻烦同学查一下词典哦！另外文章底色为灰色的句子为今天的翻译作业，今日文档中的句子分析模块会提供答案，同学们请先思考再看句子分析~】

【Para.1】

Good news on global warming is **thin on the ground**. But there may be some up in the air. Operation Blue Skies, announced on August 18th, is a scientific study being **undertaken** by Britain's air-traffic-control service, NATS, and various organisations where people study the effects on climate of the **streaks** of cloud created by aircraft—known as contrails.

- **contrail n.** /'kɒntreɪl/ (also vapor trail) a white line left in the sky by an aircraft that consists of water vapour (= gas) that has condensed (= turned back into very small drops of water) 飞机尾迹

The fighter jets left long contrails that slowly dispersed across the blue sky. 战斗机留下了长长的尾迹，在蓝天中缓慢消散。

- **thick/thin on the ground** *idiom* if people or things are thick/thin on the ground, there are a lot/not many of them in a place 为数众多（或不多）；数目巨大（或微小）；众多（或稀少）

Customers are thin on the ground at this time of year. 一年里这个季节顾客很少。

During the festival, tourists are thick on the ground in the old town. 节日期间，老城区游客密集。

- **undertake v.** /ˌʌndə'teɪk/ to make yourself responsible for sth and start doing it 承担；从事；负责

University professors both teach and undertake research. 大学教授既要教学又要从事研究工作。

v. to agree or promise that you will do sth 承诺；允诺；答应

He undertook to finish the job by Friday. 他答应星期五或之前完成这一工作。

- **streak n.** /stri:k/ a long thin mark or line that is a different colour from the surface it is on 条纹；长条痕（尤指颜色不同的）

There are these dark streaks on the surface of the moon. 月亮表面有这些黑色的条纹。

在广袤的大地上，关于全球变暖的好消息寥寥无几。但在高空之上或许还有一些转机。8月18日，英国空中交通管制服务机构 NATS 与多个气候研究机构共同开展的科学研究宣布启动，名为“蓝天行动”。研究人员专门研究飞机产生的条状云，即所谓的飞机尾迹云对气候的影响。

【Para.2】

Over the next two winters the **consortium** aims to see if it is possible to reduce the number of persistent contrails—the class with the largest such effects—over a **swathe** of the North Atlantic, thereby measurably reducing the climate **impact** of civil **aviation**. If this experiment works, a worldwide application of the same techniques might cut the climate effects of air travel by as much as half.

- **consortium** n. /kən'sɔ:tɪəm/ (pl. **consortiums** or **consortia**)
a group of people, countries, companies, etc. who are working together on a particular project (合作进行某项工程的) 财团，银团，联营企业

The consortium includes some of the biggest building contractors in North America. 该联盟包括北美一些最大的建筑承包商。

- **swathe n.** /sweɪð/ (also swath /swɒθ/) a long strip or large area especially of land 长条区域，大块区域；（尤指）长条田地，大块田地

Huge swathes of rainforest are being cleared for farming and mining. 为了耕种和采矿，大片大片的雨林被砍伐。

- **impact n.** /'ɪmpækt/ (~ of sth) (on sb/sth) the powerful effect that sth has on sb/sth 巨大影响；强大作用

The report assesses the impact of AIDS on the gay community. 这个报告评估了艾滋病对同性恋群体的影响。

v. /ɪm'pækt/ (~ on/upon sth) to have an effect on sth (对某事物) 有影响，有作用（注意重音）

The company's performance was impacted by the high value of the pound. 公司的业绩受到了英镑高值的冲击。

- **aviation n.** /ˌeɪvi'eɪʃn/ the designing, building and flying of aircraft 航空

He has worked in commercial aviation for over twenty years. 他在商业航空领域工作了二十多年。

在接下来的两个冬天，研究团队计划考察，能否通过减少北大西洋大片空域中持久性尾迹云量，从而显著降低民航对气候的影响。这类尾迹云产生的气候影响最大。如果这项实验取

得成功，再将这项技术应用到全球范围，或许可以将航空旅行对气候造成的影响降低半数之多。

【Para.3】

On a sunny day the **wispy cirrus clouds** into which some contrails are transformed can look as if they might be cooling the planet in the manner of a delicate **parasol**. And some contrails do indeed cool things down beneath them a bit. But they also warm things up by **trapping** some of the upwelling **infrared radiation** that would otherwise escape into space. On average, that warming effect, found in all contrails (and the cirrus clouds they turn into), **handsomely outweighs** the cooling provided by some. In the climate scheme of things, contrails are a liability.

- **wispy adj.** /'wɪspi/ consisting of small, thin pieces; not thick 一绺绺的；一缕缕的；成束的；纤细的

A few wispy clouds floated above the valley as the sun began to set. 夕阳西下时，几缕轻柔的云絮飘浮在山谷上方。

- **cirrus cloud n.** /'sɪrəs klaʊd/ a wispy white cloud (usually of fine ice crystals) at a high altitude (4 to 8 miles) 卷云

The sun was blazing from a blue sky feathered lightly with cirrus cloud. 阳光灿烂，蔚蓝天空中飘着些许卷云。

- **parasol n.** /'pærəsəl/ a type of light umbrella that women in the past carried to protect themselves from the sun (旧时的) 女用阳伞

Due to having weak health, she is always seen with a parasol when outside. 由于身体虚弱，她外出时总是撑着阳伞。

- **trap v.** /træp/ to prevent something from escaping, especially by capturing or enclosing it 困住；使无法逃脱

The greenhouse gases trap heat in the atmosphere. 温室气体将热量困在大气中。

n. an unpleasant situation from which it is hard to escape (难以逃脱的) 困境，牢笼

Some women see marriage as a trap. 有些妇女把婚姻视作围城。

- **infrared adj.** /,ɪnfərə'red/ having or using electromagnetic waves which are longer than those of red light in the spectrum, and which cannot be seen 红外线的；使用红外线的

The bathroom could be warmed by an infrared lamp. 可以使用红外线灯来温暖那间浴室。

- **radiation n.** /,reɪdɪ'eɪʃn/ powerful and very dangerous rays that are sent out from radioactive substances 辐射；放射线

They suffer from health problems and fear the long term effects of radiation. 他们受健康问题的困扰，还担心遭受辐射造成的长期影响。

- **handsomely adv.** /'hænsəmli/ very well or to a great degree 可观地

Their strategy paid off handsomely. 他们的策略得到了可观的回报。

- **outweigh v.** /,aʊt'wei/ to be greater or more important than sth 重于；大于；超过

The advantages far outweigh the disadvantages. 利远大于弊。

SYN exceed, surpass, prevail over

ANT fall short of

在阳光明媚的日子里，一些飞机尾迹云会逐渐转化为纤细的卷云，看起来仿佛一把精巧的遮阳伞，似乎能够为地球降温。事实上，部分尾迹云确实会让其下方的气温略有下降。但同时，它们也会通过截留一部分原本会逸散到太空中的上升红外辐射，使地球升温。平均而言，这种增温效应存在于所有尾迹云（以及它们最终转化成的卷云）中，其作用远远超过了部分尾迹云所带来的降温效果。从气候大局来看，尾迹云是一项负担。

【Para.4】

Happily, it appears that much harm can be avoided by small changes to flight paths. Persistent contrails appear only in what are known as “ice-**supersaturated** regions” (ISSRs). Often planes can avoid such regions just by flying a little bit lower. The purpose of Operation Blue Skies is to find out how well the presence of ISSRs can be predicted, how effective such little shifts in **altitude** can be in avoiding persistent contrails, and how easily and safely the changes in flight path needed to bring them about can be **integrated** into routines for airlines and air-traffic control.

- **supersaturated** **adj.** /ˌsju:pə'sætʃəreɪtɪd/ containing more of a substance than can be absorbed 超饱和的

Meteorologists are studying how supersaturated air masses trigger the formation of ice crystals. 气象学家正在研究过饱和气团如何触发冰晶的形成。

- **altitude** **n.** /'æltɪtju:d/ the height above sea level 海拔；海拔高度；高程

We are flying at an altitude of 6 000 metres. 我们的飞行高度是 6000 米。

(Words of Similar Form) **attitude** **n.** /'ætɪtju:d/ (~to/towards sb/sth) the way that you think and feel about

sb/sth; the way that you behave towards sb/sth that shows how you think and feel 态度；看法

Being unemployed produces negative attitudes to work. 失业会对工作产生消极态度。

- **integrate v.** /'ɪntɪɡreɪt/ (~ A) (into/with B) (~ A and B) to combine two or more things so that they work together; to combine with sth else in this way (使) 合并，成为一体

These programs will integrate with your existing software. 这些程序将和你的已有软件可综合成整体。

好在，似乎只需对飞行航线做些小幅调整，就能避免其中相当一部分危害。持久性尾迹云只会出现在所谓的“冰过饱和区”内。飞机往往只需稍微降低飞行高度，就可以避开这些区域。“蓝天行动”的目的，就是弄清楚以下几个问题：ISSR 的存在能否被准确预测；如此小幅的高度调整在多大程度上能够有效避免持久性尾迹云；以及，为实现这一目的所需的航线调整，如何能以便捷且安全的方式纳入航空公司和空中交通管制的日常运行流程。

【Para.5】

The **cut** of up to half in the climate-effects of air travel would come at a very small cost. The number of flights at lowered altitude might be small—as little as 5% of the total. Because altitude **buys** efficiency, those

flights would consume more jet fuel, but less than 2% more.

- **cut n.** /kʌt/ (~ in sth) a reduction in amount, size, supply, etc. (数量、尺寸、供应等的) 削减, 减少, 缩减

They had to take a 20% cut in pay. 他们不得不接受减薪 20%。

n. a share in sth, especially money (尤指钱的) 份, 份额

They were rewarded with a cut of 5% from the profits. 他们得到了占利润 5% 份额的酬报。

a cut above sb/sth idiom better than sb/sth 优于; 比...高一等; 胜.....一筹

His latest novel is a cut above the rest. 他最近出版的小说比其他的小小说都好。

- **buy v.** /baɪ/ if you buy something like time, freedom, or victory, you obtain it but only by offering or giving up something in return 换取

It was a risky operation, but might buy more time. 这是一项冒险的举动, 但也许能换得更多的时间。

v. to persuade sb to do sth dishonest in return for money 买通; 收买; 贿赂

He can't be bought. 他是收买不了的。

v. to obtain sth by losing sth else of great value 付出极大的代价而获得

Her fame was bought at the expense of her marriage. 她出了名，却牺牲了她的婚姻。

将航班对气候的影响减少半数之多，所付出的代价非常小。需要降低飞行高度的航班可能并不多，仅占航班总量的 5%。由于飞行高度越高，燃油效率越高，这些降低飞行高度的航班会消耗更多航空燃油，但增加量不到 2%。

【Para.6】

Burning even that little extra fuel would be **anathema** to airlines. So some suggest the costs of action might be minimised, and the airlines **incentivised**, by setting up a market which somehow links contrail cooling to carbon prices. The European Union, which has had some success and plenty of experience with carbon markets, recently took a small step towards something along these lines. It is also the sort of idea to which economists—and The Economist—tend to be **partial**. But it would be a mistake nonetheless.

- **anathema n.** /ə'næθəmə/ (formal) a thing or an idea which you hate because it is the opposite of what you believe 可憎的事物；可恶的想法

For older employees, the new system is an anathema. 对于年纪较大的职员来说，新制度就是眼中钉，肉中刺。

- **incentivise v.** /In'sentivaɪz/ (also incentivize) to make someone want to do something 刺激，激励

We need to incentivize our sales managers to achieve these targets. 我们需要激励我们的销售经理实现这些目标。

- **partial adj.** /'pɑ:ʃl/ having a liking for something 偏爱...的，喜欢...的

I'm not partial to fish. 我并不特别爱吃鱼。

adj. not complete or whole 部分的；不完全的

The general has ordered a partial withdrawal of troops from the area. 将军已下令部队部分人员撤出该地区。

哪怕只是多烧这么一点点燃油，也是航空公司极其反感的。因此，有人提出，可以建立一种将尾迹云带来的降温效应与碳价格联系起来的机制，以尽可能降低采取相关措施的成本，同时为航空公司提供激励。欧盟在碳市场方面不仅取得了一定成效，也积累了丰富经验，最近已朝着这一方向迈出了一小步。这也是经济学家，包括《经济学人》在内，往往比较偏爱的思路之一。但即便如此，这样做仍然是错误的。

【Para.7】

Markets for pollutants have worked best when both the aim and the remedy are fairly simple—most famously, reducing **sulphur-dioxide emissions** from power stations in America. Contrails, by contrast, are **complicated**. Though there is no doubt that their net effect is warming, saying how much warming a particular contrail might produce is often difficult and sometimes next to impossible.

- **sulphur n.** /'sʌlfə(r)/ (NAme sulfur) (symbol S) a chemical element. Sulphur is a pale yellow substance that produces a strong unpleasant smell when it burns and is used in medicine and industry. 硫；硫磺

The laboratory assistant detected a faint smell of burning sulphur after the experiment. 实验助理在实验后闻到了轻微的硫磺燃烧气味。

- **dioxide n.** /daɪ'ɒksaɪd/ a substance formed by combining two atoms of oxygen and one atom of another chemical element 氧化物

Plants absorb carbon dioxide from the atmosphere during photosynthesis. 植物在光合作用过程中从大气中吸收二氧化碳。

- **emission n.** /i'mɪʃn/ (formal) the production or sending out of light, heat, gas, etc. (光、热、气等的) 发出，射出，排放

The emission of gases such as carbon dioxide should be stabilized at their present level. 二氧化碳之类气体的排放应该被控制在目前的水平上。

- **complicated adj.** /'kɒmplɪkətɪd/ made of many different things or parts that are connected; difficult to understand 复杂的；难懂的

The situation in Lebanon is very complicated. 黎巴嫩的形势非常复杂。

SYN complex, intricate

在目标明确，而且解决办法相对简单时，污染物交易市场运转得最为成功。最著名的例子是，美国减少发电厂二氧化硫排放的做法。相比之下，尾迹云要复杂得多。尽管毫无疑问，尾迹云总体上会产生增温效应，但要判断某一片特定的尾迹云究竟可能造成多大程度的增温，往往非常困难，有时甚至几乎不可能。

【Para.8】

The difficulties are made **yet** worse when the contrail being **assessed** is a purely notional one—as would be the case for the contrails avoided by changing flight paths. Valuing **counterfactuals** is always hard.

- **yet adv.** /jet/ (~ worse, more importantly, etc.) used to emphasize an increase in the degree of sth (= how bad, important, etc. it is) (强调程度的增加) 更

The new software is fast, but the upgraded version is yet faster. 新软件很快，但升级版甚至更快。

adv. (could, might, may, etc. do sth ~) used to say that sth could, might, etc. happen in the future, even though it seems unlikely (表示将来可能发生，尽管现在似乎没有可能) 早晚，总有一天

She could yet surprise us all. 总有一天，她会让我们都大吃一惊。

adv. now; as soon as this 现在；即刻；马上

We don' t need to start yet. 我们不必马上开始。

adv. from now until the period of time mentioned has passed 从现在起直至某一时间；还

They won' t arrive for at least two hours yet. 他们至少要过两个小时才能到。

- **assess v.** /ə'ses/ (~ sb/sth as sth) to make a judgement about the nature or quality of sb/sth 评估，评定 (性质、质量)

It's difficult to assess the effects of these changes. 这些变化带来的效果难以评估。

前缀：as-朝，向，加强

assist v. /ə'sist/ 帮助 (as- 朝，向，加强 + sist 站立)

assure v. /ə'ʃʊə(r)/ 保证 (as- 朝，向，加强 + sure 肯定)

assimilate v. /ə'siməleɪt/ (使)同化 (as- 朝，向，加强 + simil- 类似，一样 + -ate 做，造成)

- **counterfactual n.** /ˌkaʊntə'fæktʃuəl/ expressing what has not happened but could, would, or might under differing conditions 反事实的

Thoughts about how an embarrassing event might have turned out differently are known to psychologists as counterfactual thinking. 心理学家把人们想象已发生的尴尬事件其实可能以不同方式发生这样的想法叫做违实思考。

如果被评估的尾迹云只存在于假设当中，比如可以通过改变飞行航线避免的尾迹云，问题就会更加棘手。对反事实情形进行估值，向来很困难。

【Para.9】

If contrail-avoidance turns out to be as useful as it looks, it would be much simpler to simply have air-traffic controllers **assign** routes that limit

persistent contrails when it is possible and safe to do so. If some airlines choose to avoid the affected routes, others will surely be happy to take their place. And if such a **scheme** were also to show the benefits of innovations in air-traffic control (quite a few have been suggested) the industry would end up **better off**, as well as less polluting.

- **assign v.** /ə'sain/ (~ sth to sb) (~ sb sth) to give sb sth that they can use, or some work or responsibility 分配 (某物); 分派, 布置 (工作、任务等)

The two large classrooms have been assigned to us. 这两间大教室分配给了我们。

The teacher assigned a different task to each of the children. 老师给每个儿童布置的作业都不相同。

- v.** (~ sb to sth/as sth) to provide a person for a particular task or position 指定; 指派

They' ve assigned their best man to the job. 他们指派了最优秀的人担任这项工作。

- **scheme n.** /ski:m/ a plan or system for doing or organizing sth 计划; 方案; 体系; 体制

Under the new scheme only successful schools will be given extra funding. 在新体制下, 只有办得好的学校才可获得额外经费。

● **well off adj.** rich 富有的，有钱的

a well-off neighbourhood 富人区

adj. having a lot of or a number of 拥有大量...的；充足的

The city is well off for parks and gardens. 这座城市有很多公园和花园。

如果规避尾迹云的做法确实像目前看起来那样有效，那么最简单的办法就是：只要条件允许且安全可行，就由空中交通管制员安排能够减少持久性尾迹云的航线。如果部分航空公司选择避开受影响的航线，其他航空公司肯定会乐于填补空缺。而且，如果这种方案还能展示空中交通管制创新所带来的好处（目前已经有人提出了不少相关方案），那么最终整个航空业不仅会减少污染，也会从中获益。

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今日翻译作业

(Para. 2) Over the next two winters the consortium aims to see if it is possible to reduce the number of persistent contrails—the class with the largest such effects—over a swathe of the North Atlantic, thereby measurably reducing the climate impact of civil aviation. （下一页提供句子分析）

今日句子分析

Para. 2 Over the next two winters the consortium aims to see if it is possible to reduce the number of persistent contrails—the class with the largest such effects—over a swathe of the North Atlantic, thereby measurably reducing the climate impact of civil aviation. 在接下来的两个冬天，研究团队计划考察，能否通过减少北大西洋大片空域中持久性尾迹云量，从而显著降低民航对气候的影响。这类尾迹云产生的气候影响最大。

主句

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时间状语

the consortium

主语

aims to see

谓语

宾语从句

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同位语

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后置定语

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地点状语

thereby measurably reducing the climate impact

结果状语

of civil aviation.

后置定语

主句：

时间状语：Over the next two winters；

主语：the consortium；谓语：aims to see；

宾语从句：

引导词：if；形式主语：it；系语：is；表语：possible；真正的主语：to reduce；

宾语：the number of persistent contrails；

同位语：the class；后置定语：with the largest such effects；（进一步说明 persistent contrails 是什么性质的尾迹云，可理解为“持久性尾迹云，即其中气候影响最大的一类”。这里的 class 表示“类别、类型”，such 是回指，指代前文刚刚提到的性质。

译文重新调整为“这类尾迹云产生的气候影响最大”，汉语使用主谓结构更加自然。由于这部分内容仅为补充说明，较为次要，因此放在句子最后，优先保证主干的连贯性。）

地点状语：over a swathe of the North Atlantic；（over 表示在“在……上空 / 覆盖……范围”，没有搭配动词，进一步说明这些尾

迹云出现在哪里。)

结果状语: thereby measurably reducing the climate impact;

(thereby 表示“由此, 从而, 因此”, 语言风格很正式, 常见于学术论文、法律文本、正式报告等, 口语中很少用到。后加 V-ing, 这就是 reducing 的由来。注意区分它和 therefore, therefore 强调的是逻辑结论: The weather was bad. Therefore, the flight was cancelled. (天气不好, 因此航班取消了。)) thereby 强调的是通过某种动作/手段, 从而产生某个结果: The airline changed the flight path, thereby reducing emissions. (航空公司调整了航线, 从而减少了排放。) 本文就是这个逻辑, 所以译文增译了“通过”, 把原文隐含的“手段—结果”关系显化。)

后置定语: of civil aviation;

写作积累

1. (Para. 1) But there may be some **up in the air**.

本句中的 up in the air 表示 “high in the clouds” 的同时，也表示 “uncertain, often because other matters have to be decided first”，即 “尚未有定论，悬而未决，尚未决定”，例如，Our travel plans are still up in the air. 我们的旅行计划尚未决定。这里是原作者设置的双关。此外，还有以下习语可以表示相类似的含义：

- ♦ **up for grabs** *idiom* something is available for anyone to take, win, or claim 待价而沽，谁都能拿，尚未定论
The ad says, “ World’ s Greatest Job, Up for Grabs. ” 广告上说：“世界上最棒的工作，虚位以待。”
- ♦ **on the fence** *idiom* unable to decide between two options or refuse to take a side in an argument 犹豫不决，左右为难
The committee is still on the fence about the proposal. 委员会对这项提案仍然犹豫不决。

2. (Para. 4) The purpose of Operation Blue Skies is to find out how well the presence of ISSRs can be predicted, how effective such little shifts in altitude can be in avoiding persistent contrails, and how easily and safely the changes in flight path needed to **bring** them **about** can be

integrated into routines for airlines and air-traffic control.

本句中的 bring about 表示 “to cause something to happen or make a change occur”，即 “带来，引起，实现”，例如，The new policy brought about significant changes in the economy. 新政策引起了经济上的重大变化。此外，还有以下习语可以表示相同的含义，供写作时选用：

- ♦ **give rise to** *idiom* to cause something to happen or to be the origin of a result 引起

The new policy will give rise to much discussion. 这项新政策将引起很多讨论。

- ♦ **set off** *idiom* cause something to start 触发，引发

The announcement set off a wave of protests. 这项声明引发了一波抗议。